

Code No:- V3121/R07

Set No: 1

III B.Tech I Semester Supplementary Examinations, April/May - 2012

DIGITAL IC APPLICATION

(Electronics and Communications Engineering)

Time: 3 Hours**Max Marks: 80**

Answer any FIVE Questions
All Questions carry equal marks

1. a) Draw the resistive model of a CMOS inverter and explain its behavior for LOW and HIGH outputs.
b) Explain the following terms?
i) Logic levels ii) D.C noise margin iii) Power supply rails iv) Propagation delay [8+8]
2. a) List out different categories of characteristics in a TTL data sheet. Discuss electrical and switching characteristics of 74LS00?
b) Design a transistor circuit of 2 input ECL NOR gate. Explain the operation with the help of function table. [8+8]
3. a) Write a VHDL Entity and Architecture for a 3-bit synchronous counter using Flip-Flops.
b) Explain the use of Packages. Give the syntax and structure of a package in VHDL. [8+8]
4. Design the logic circuit and write a data-flow chart style VHDL program for the following function.
a) $F(A) = \pi_{p,q,r,s} (1,3,4,5,6,7,9,12,13,14)$
b) $F(X) = \sum_{A,B,C,D} (3,5,6,7,13) + d(1,2,4,12,15)$ [8+8]
5. a) Draw the logic symbol of 74X245 and explain its operation?
b) Write a data flow style VHDL program for 74X245 [8+8]
6. a) Design an Excess-3 decimal counter using 74X163 and explain the operation with the help of timing waveforms.
b) Design a modulo – 100 counter using two 74X163 binary counters. [8+8]
7. a) Differentiate between ripple counter and synchronous counter? Design a 4-bit counter in both modes and estimate the propagation delay.
b) Design a modulo-88 counter using 74X163 Ics. [8+8]
8. a) Draw the block diagram of synchronous SRAM and explain each one of it precisely.
b) Explain the internal structure of 64KX1 DRAM .With the help of timing waveforms. Describe the DRAM access. [8+8]

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Set No: 2

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1. a) Design a 4 input CMOS OR-AND-INVERT gate. Explain the circuit with the help of logic diagram and function table.
b) Compare different logic families? [8+8]
2. a) What is the necessity of separate interfacing circuit to connect CMOS gate to TTL gate? Draw the interface circuit and explain the operation.
b) Design a TTL three input NAND gate with open collector output and explain the operation with the help of function table. [8+8]
3. a) Discuss the steps in VHDL design flow.
b) Write a VHDL Entity and Architecture for the following function.
 $F(x) = (a+b)(c+d)(e+f)$
Also draw the relevant logic diagram. [8+8]
4. a) Design 8 bit comparator using two 74283 IC.
b) Design a logic circuits write structural style VHDL program for the following function
 $F(A) = \sum_{P,Q,R,S}(1,3,4,5,6,7,9,12,13,14)$ [8+8]
5. a) Design a 32 to 1 multiplexer using four 74X151 multiplexers and 74X139 decoder.
b) Realize the following expression using 74X151 IC
 $F(Y) = AB + BC + AC$ [8+8]
6. a).Design a barrel shifter for 8 bit using three control inputs? Write a VHDL program for the same in dataflow style
b) Write a behavioral VHDL program to compare 16 bit signed and unsigned integers. [8+8]
7. a) Design a 8 bit parallel-in and serial-out shift register. Explain the operation of the above Shift register with the help of timing waveforms.
b) Design a 3 bit LFSR counter using 74x194. List out the sequence assuming that the Initial state is 101. [8+8]
8. a) Design an 8x4 diode ROM using 74X138 for the following data starting from the first location. (1, 4, 9, B, A, 0, F, C)
b) Draw the internal structure of synchronous SRAM and explain its operation? [8+8]

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Set No: 3

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DIGITAL IC APPLICATION

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Time: 3 Hours**Max Marks: 80**

Answer any FIVE Questions
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1. a) Explain how to estimate sinking current for low output and sourcing current for high output of CMOS gate.
b) Analyze the fall time of CMOS inverter output with $R_L = 100\Omega$, $V_L = 2.5V$ & $C_L = 10PF$. Assume V_L as stable state voltage. [8+8]
2. a) Design a TTL three-input NAND gate with totem pole output and explain the operation with the help of function table.
b) Compute the maximum fan-out for the following cases.
i) 74LS driving 74F
ii) 74F driving 74AS
iii) 74AS driving 74LS
iv) 74AS driving 74F [8+8]
3. a) Explain with an example the syntax and the function of the following VHDL statements
i) Process statement ii) case statement
b) Explain implicit and explicit visibility of a library in VHDL [8+8]
4. a) Design a logic circuit to detect prime number of a 4-bit input? Write the VHDL program for the above design?
b) Design the logic circuit and write a data-flow style VHDL program for the following function?
 $F(P) = \sum_{A,B,C,D} (1, 5, 6, 7, 9, 13) + d(4, 15)$ [8+8]
5. a) Design a full subtractor with logic gates and write VHDL data flow program for the implementation of the above subtractor.
b) Using the above subtractor design a 8-bit ripple subtractor and write the corresponding VHDL program in structural style of modeling. [8+8]
6. Write VHDL program for 8-bit comparator circuit. Using this entity write VHDL program for 24-bit comparator. Show the additional logic used for this purpose use structural style? [16]
7. a) Design a three bit LFSR counter using 74X194. List out the sequence assuming that the initial state is 110?
b) Design an 8-bit serial in and serial out shift register. Write the data flow VHDL program for this shift register. [8+8]
8. a) Describe DRAM with an appropriate diagram and explain about its timings.
b) Distinguish between PROM, EPROM and EEPROM. [8+8]

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Code No:- V3121/R07**Set No: 4**

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1. a) Design a CMOS transistor circuit with the functional behavior $f(x) = (A+B)(C+D)$
b) Distinguish between static and dynamic power dissipation of a CMOS circuit. Derive the expression for dynamic power dissipation [8+8]
2. a) Design a transistor circuit of 2 input ECL NOR gate. Explain the operation with the help of function table.
b) List out TTL families and compare them with reference to propagation delay, power consumption, speed power product and low level input current [8+8]
3. a) Explain with an example the syntax and the function of the following VHDL statements.
i) Loop statements
ii) if, else, and else if statements
b) Write a VHDL entity and Architecture for 3-bit ripple counter using Flip-Flops [8+8]
4. Design a logic circuit to detect prime number of a 5-bit input. Write the structural VHDL program for the same. [16]
5. a) Write a data flow style VHDL program for a simple 8 bit multiplexer.
b) Design a priority encoder for 16 inputs using two 74×148 encoders. [8+8]
6. A simple floating-point encoder converts 16-bit fixed-point data using four high order bits beginning with MSB. Design the logic circuit and write VHDL data-flow program. [16]
7. a) Design an Excess-3 decimal counter using 74×163 and explain the operation with help of timing waveforms.
b) It is necessary to generate 6 control lines in regular intervals sequentially. Design the necessary circuit using 74×163 and 74×138. [8+8]
8. a) Explain the internal structure of 64K×1 DRAM. With the help of timing waveforms discuss DRAM access.
b) Design an 8×4 diode ROM using 74×138 for the following data starting from the first location. (F, 2, 9, C, E, 1, F, A) [8+8]
